Do not write outside the box

Please write clearly in block capitals.

Centre number | | | | | |

Candidate number | | | | |

Surname

Forename(s)

Candidate signature

Combined Science: Trilogy

Biology Paper 1 topics (4.1 to 4.4)

Foundation Tier

Suggested time: about 30 minutes. There is no timer.

Materials

For this paper you must have:

- a ruler
- a scientific calculator

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer all questions in the spaces provided.
- Do all rough work in this book. Cross through any work you do not want to be marked.
- In all calculations, show clearly how you work out your answer.

Information

- The maximum mark for this paper is 30.
- The marks for questions are shown in brackets.
- You are expected to use good English and clear presentation in your answers.

For Examiner's Use	
Question	**Mark**
1	
2	
3	
4	
5	
6	
TOTAL	

This is a practice paper written for Dougie in the style of the GCSE exam. It is not an exam board paper, and it is not a paper from any past exam series.

0 1 . 1 Which structure is found in a plant cell but not in an animal cell? Tick (✓) one box.

Cell membrane ☐

Chloroplast ☐

Mitochondrion ☐

Ribosome ☐

[1 mark]

0 1 . 2 A student looks at a cheek cell through a light microscope. The image of the cell is 6 mm wide. The real cell is 0.06 mm wide. Calculate the magnification. Show your working.

Magnification = × .....

[2 marks]

0 1 . 3 Before a cell divides by mitosis, it goes through a stage of growth. Describe two things that happen to the cell during this stage.

[2 marks]

0 2 A student put cylinders of potato into sugar solutions of different concentrations. She weighed each cylinder before and after leaving it for 30 minutes.

Concentration of sugar solution in mol/dm³	Percentage change in mass
0.0	+12.0
0.2	+4.5
0.4	-2.0
0.6	-8.5
0.8	-12.5

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0 2 . 1 One cylinder had a mass of 2.50 g at the start and 2.80 g at the end. Calculate the percentage change in mass. Show your working.

Percentage change = ... %

[2 marks]

0 2 . 2 Between which two concentrations in the table would a cylinder show no change in mass?

Between ..... and ..... mol/dm³

[1 mark]

0 2 . 3 Explain why the cylinder in the 0.8 mol/dm³ solution lost mass.

[3 marks]

0 2 . 4 Why did the student calculate the percentage change in mass, rather than just the change in mass?

[1 mark]

0 3 . 1 Name the enzyme that breaks down lipids, and name the products.

[2 marks]

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0 3 . 2 Explain why an enzyme stops working at a high temperature.

[2 marks]

0 3 . 3 Name the reagent used to test for protein, and give the colour it turns if protein is present.

[2 marks]

0 4 . 1 Explain how a stent helps a person with coronary heart disease.

[2 marks]

0 5 . 1 Measles is caused by a virus. Explain why a doctor would not give antibiotics to treat measles.

[1 mark]

0 5 . 2 Explain how a vaccination can stop a person becoming ill with measles.

[3 marks]

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0 6 . 1 A student wants to find out how light intensity affects the rate of photosynthesis in pondweed. Describe a method the student could use. Include how they would make it a fair test.

[6 marks]

END OF QUESTIONS

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